# Genomic & Pharmacogenomic Clinical Summary

**Patient:** Melinda Ehrle (Melinda_Ehrle) | **Date:** October 13, 2026

## 1. PRIORITY GENOMIC FINDINGS & CARRIER STATUS

| GENE | VARIANT | ZYGOSITY | CLASSIFICATION | CLINICAL IMPACT & GUIDANCE |
|---|---|---|---|---|
| **POLG** | p.Gly737Arg · c.2209G>C · rs121918054 | Heterozygous | Pathogenic | Strict contraindication against valproic acid due to fatal liver toxicity risk. Carrier retains normal baseline mitochondrial function. |
| **TAT** | p.Arg57Ter · c.169C>T · rs118203914 | Heterozygous | Pathogenic | Single-copy heterozygous carrier; maintains normal enzymatic function on standard diet. No protein or tyrosine restriction required. |
| **ATM** | c.1236-2del · rs1565381646 | Heterozygous | Pathogenic | Moderately elevated cancer susceptibility. Discuss guideline-directed breast/preventive screening (annual MRI/mammogram) with PCP. |
| **VDR** | c.-1172A>G · rs4516035 | Heterozygous | Pathogenic | Modulates calcitriol receptor signaling. Check routine 25(OH)D at annual checkup to guide standard vitamin D3 supplementation. |
| **MC1R** | p.Asp294His · c.880G>C · rs1805009 | Heterozygous | Risk Factor | Increased UV skin sensitivity. Annual full-body skin examination and routine broad-spectrum SPF 30+ sun protection. |
| **AMPD1** | p.Glu435Gly · c.1304A>G · rs764950035 | Heterozygous | VUS | Single-copy heterozygous carrier; normal muscle energy metabolism. Symptom-guided check only if exercise cramping or statin myalgia occurs. |
| **CTH** | p.Thr67Ile · c.200C>T · rs28941785 | Heterozygous | Modifier | Preserves baseline homocysteine clearance. Review routine B-vitamin intake (B6, B12, folate) during annual wellness checks. |
| **PLK4** | p.Gly6Glu · c.17G>A · rs149003893 | Heterozygous | Modifier | Potential impact on Inborn genetic diseases pathway. Single-copy heterozygous finding; review in the context of standard annual wellness labs and personal clinical history. |
| **TMEM43** | p.Trp316Ser · c.947G>C · rs199526104 | Heterozygous | Modifier | Single-copy heterozygous carrier maintains normal adult hearing. Consider baseline hearing check; avoid unnecessary ototoxic aminoglycosides. |
| **SUMF1** | p.Asn352Lys · c.1056C>G · rs376126385 | Heterozygous | VUS | Potential impact on Multiple sulfatase deficiency pathway. Single-copy heterozygous finding; review in the context of standard annual wellness labs and personal clinical history. |
| **CHST14** | p.Gln349Pro · c.1046A>C · rs1463514561 | Heterozygous | VUS | Potential impact on Cardiovascular phenotype. Single-copy heterozygous finding; review in the context of standard annual wellness labs and personal clinical history. |
| **NPC2** | c.441+1G>A · rs140130028 | Heterozygous | Modifier | Modulates lipoprotein clearance. Review routine fasting lipid panel and cardiovascular wellness goals at annual checkup. |

# Prescribing Precautions & Annual Checkup Discussion Topics

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## 2. HIGH-PRIORITY MEDICATION & EXPOSURE PRECAUTIONS (CPIC / FDA)

| GENE & DIPLOTYPE | PHENOTYPE | AFFECTED MEDICATIONS | PRESCRIBING ACTION & GUIDANCE |
|---|---|---|---|
| **POLG** p.Gly737Arg · c.2209G>C · rs121918054 | Mitochondrial Carrier | ⚠️ LIFE-THREATENING RISK **Valproic Acid / Sodium Valproate** | **ABSOLUTE CONTRAINDICATION (FATAL RISK):** High risk of fatal valproate-induced hepatic necrosis (FDA Boxed Warning / CPIC). Strictly avoid valproate formulations. |
| **ABCG2** *ref/*rs2231142 (Q141K (rs2231142) G>T) | Decreased Function | DOSE_DOWN Allopurinol, Clozapine, Imatinib, Methotrexate | **DOSE_DOWN:** Allele G is associated with decreased dose of allopurinol or febuxostat in people with Gout as compared to allele A. |
| **CYP2C19** *2/*2 (splice defect (rs12769205) A>G, splice defect (rs4244285) G>A) | Poor Metabolizer | ⚠️ HIGH RISK · AVOID **Clopidogrel (Plavix)** | **AVOID / CONTRAINDICATION:** Significantly reduced active metabolite formation and high risk of cardiovascular ischemic events/stent thrombosis. Avoid clopidogrel; prescribe alternative antiplatelet agent (Prasugrel or Ticagrelor). |
| **CYP3A43** *2/*3 (Y25fs (rs61469810) delA, P340A (rs680055) C>G) | Unknown | MONITOR CPIC Substrates | **MONITOR:** Titrate dosing per CPIC. |
| **CYP3A5** *3/*3 (splice defect (rs776746) T>C) | Poor Metabolizer | STANDARD Tacrolimus (Prograf) | **STANDARD:** CYP3A5 *3/*3 non-expressor (Poor Metabolizer). Requires standard lower initial dosing (0.10-0.15 mg/kg/day) compared to CYP3A5 expressors to achieve target trough concentrations. Monitor blood levels (TDM). |

## 3. HIGH-LEVEL PREVENTIVE CARE & ANNUAL CHECKUP TOPICS

| PREVENTIVE CARE TOPIC | TRIGGERING GENE & VARIANT | TIMING | CLINICAL CONTEXT & GUIDANCE |
|---|---|---|---|
| **EHR Valproate Allergy/Avoidance Flag & Routine Liver Panel Permanent Prescribing Precaution** | POLG p.Gly737Arg · c.2209G>C · rs121918054 Heterozygous (Autosomal Dominant) | Permanent EHR Flag & Annual Checkup | Strict contraindication against valproic acid due to fatal liver toxicity risk. Carrier retains normal baseline mitochondrial function. |
| **Reassurance Only — Normal Unrestricted Diet Optional Routine Metabolic Panel at Annual Checkup** | TAT p.Arg57Ter · c.169C>T · rs118203914 Heterozygous (Autosomal recessive) | Routine Annual Checkup | Single-copy heterozygous carrier; maintains normal enzymatic function on standard diet. No protein or tyrosine restriction required. |
| **Guideline-Directed Breast & Cancer Screening Discussion (per NCCN) Preventive Screening Review** | ATM c.1236-2del · rs1565381646 Heterozygous (Autosomal recessive) | Annual Preventive Screening Review | Moderately elevated cancer susceptibility. Discuss guideline-directed breast/preventive screening (annual MRI/mammogram) with PCP. |
| **Routine Vitamin D & Calcium Check Annual Wellness Panel** | VDR c.-1172A>G · rs4516035 Heterozygous (Pathway Modifier) | Annual PCP Checkup | Modulates calcitriol receptor signaling. Check routine 25(OH)D at annual checkup to guide standard vitamin D3 supplementation. |
| **Annual Skin Examination & Daily Sun Protection Dermatology / PCP Skin Check** | MC1R p.Asp294His · c.880G>C · rs1805009 Heterozygous (Pathway Modifier) | Annual Preventive Skin Check | Increased UV skin sensitivity. Annual full-body skin examination and routine broad-spectrum SPF 30+ sun protection. |

# Clinical Perspective, Evidence Context & EHR Chart Summary

**Patient:** `Melinda Ehrle` (`Melinda_Ehrle`)

## 4. CLINICAL PERSPECTIVE: PROACTIVE CARE VS. AVOIDING OVER-INTERVENTION

### ✅ WHY THESE FINDINGS ARE USEFUL

- **Proactive Medication Safety:** Documenting pharmacogenomic diplotypes prevents avoidable drug side effects and subtherapeutic dosing before a prescription is started.
- **Routine Preventive Care:** High-level wellness topics integrate naturally into the annual primary care physical.
- **Specialist Coordination:** Sharing finding-specific precautions supports informed care without unnecessary medicalization.

### 🛑 WHY AGGRESSIVE INTERVENTION IS NOT NEEDED

- **Single-Copy Carrier Reassurance:** Single-copy recessive carriers retain a functional gene copy and maintain normal baseline health.
- **Absence of Biallelic Recessive Disease:** Zero two-copy pathogenic recessive disease genotypes are present in this profile.
- **Pathway Modifiers Are Not Diagnoses:** Common physiological modifiers should never trigger unprovoked medication changes or invasive testing.

## 5. CLINICAL EVIDENCE CONTEXT BY FINDING CATEGORY

| FINDING CATEGORY | EVIDENCE TIER | CLINICAL INTERPRETATION BASIS |
|---|---|---|
| **Primary Carrier & Susceptibility Findings (POLG, TAT, ATM)** | High | Established clinical curation records (ClinVar / OMIM) with clear inheritance context. |
| **Pharmacogenomic Diplotypes (ABCG2, CYP2C19, CYP3A43)** | High | CPIC / DPWG clinical dosing guidelines with established star-allele assignments. |
| **Physiological & Pathway Modifiers** | Moderate | Population-level associations guiding general wellness discussions rather than diagnostic labeling. |

## 6. COPY-PASTE PHYSICIAN HANDOFF / EHR CHART SUMMARY

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CLINICAL GENOMICS & PHARMACOGENOMICS SUMMARY (Melinda Ehrle | ID: Melinda_Ehrle):
- PRIORITY GENOMIC & CARRIER FINDINGS: POLG (p.Gly737Arg · c.2209G>C · rs121918054, Heterozygous); TAT (p.Arg57Ter · c.169C>T · rs118203914, Heterozygous); ATM (c.1236-2del · rs1565381646, Heterozygous); VDR (c.-1172A>G · rs4516035, Heterozygous); MC1R (p.Asp294His · c.880G>C · rs1805009, Heterozygous). Single-copy recessive carrier findings retain one functional gene copy and do not cause primary recessive disease.
- PHARMACOGENOMIC PRECAUTIONS (CPIC / FDA): ABCG2 *ref/*rs2231142. Review pharmacogenomic diplotypes prior to prescribing substrate medications.
- PREVENTIVE CARE PLAN: Discuss high-level wellness and screening topics (Section 3) during routine primary care checkups.
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